MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Value Addition Course (VAC) - MG3VACAQC200
AQUATIC POLLUTION AND MANAGEMENT
Question paper code :25900707

Part A
Very Short Answer Questions
Answer all questions
Each question carries 1 marks (1x10= 10)

1. Name any two major sources of aquatic pollution.
Industrial discharge, agricultural run off, heavy metals, sewage, domestic waste, oil, radioactive substances, hot water etc.
2. Name any two pollutants that belong to the category of agricultural waste.
Pesticides, fertilizers
3. Write two examples for biodegradable waste.
Vegetable peel, animal dung, fruit peel
4. Name one source of nitrogen in aquatic ecosystems.
Agricultural runoff, Sewage and animal waste, Atmospheric deposition, Decomposition of organic matter
5. Write a source that contribute iron in aquatic ecosystem
Weathering of rocks and soil , Industrial effluents, Runoff from agricultural lands, Sewage and domestic waste, Atmospheric deposition
6. What is the objective of The Environment (Protection) Act?
To protect and improve the quality of the environment and to prevent, control, and abate environmental pollution.
7. What happens to submerged vegetation during algal bloom?
Sunlight is blocked, Reduced photosynthesis, Oxygen depletion
8. Which index categorizes organisms by their tolerance to organic pollution?
Saprobic index or biotic index
9. Define microbial pollution
Microbial pollution refers to the contamination of water by pathogenic microorganisms such as bacteria, viruses, protozoa, and fungi, which originate mainly from sewage, animal waste, or untreated wastewater.
10. Give one example of a toxic organic compound found in polluted water.
Polychlorinated biphenyls (PCBs), Pesticides, Polycyclic aromatic hydrocarbons (PAHs), Phenols

Part B
Short Answer Questions
Answer any 10 questions
Each question carries 2 marks (2X10=20)

11. What are the major impacts of thermal pollution in aquatic environment?
Reduced dissolved oxygen, Altered metabolism of aquatic organisms, Disruption of reproductive cycles, Increased susceptibility to diseases, Loss of biodiversity
12. Explain how industrial effluents pollute aquatic ecosystems.

Chemical contamination, Thermal pollution, Eutrophication, Introduction of hazardous organic compounds, Sedimentation and turbidity. Overall, industrial effluents alter water chemistry, harm organisms, and disrupt ecosystem balance.

13. Differentiate between natural and man-made aquatic pollution.
Natural aquatic pollution : Occurs from natural processes like soil erosion, volcanic eruptions, and decaying vegetation, Mostly sediments, organic matter, and minerals.
Often occasional or seasonal, Usually limited and part of natural ecosystem cycles.
Man-made aquatic pollution : Caused by human activities such as industrial discharge, agricultural runoff, and sewage disposal, Chemicals, heavy metals, pesticides, fertilizers, plastics, and pathogens, Continuous or frequent, depending on human activity, Often severe, persistent, and harmful to aquatic life and human health
14. Why is bioremediation considered a natural process?
Bioremediation is considered a natural process because it uses living organisms—such as bacteria, fungi, and plants—to break down, degrade, or transform pollutants in the environment into less harmful substances. The process mimics natural decomposition and metabolic activities that occur in ecosystems, accelerating the self-purification of soil and water without introducing synthetic chemicals.
15. Define Chemical Oxygen Demand (COD) and explain why it is important.
Chemical Oxygen Demand (COD) is the amount of oxygen required to chemically oxidize the organic and inorganic matter present in a water sample using a strong chemical oxidant, usually potassium dichromate, under acidic conditions. It is expressed in milligrams of oxygen per liter (mg/L).
Importance of COD: Measures pollution level, Quick assessment, Industrial monitoring, Ecosystem health
16. Comment on the impact of toxic organic compounds on aquatic organisms
Direct toxicity, Bioaccumulation, Disruption of reproduction and, Altered behavior and physiology, Ecosystem imbalance
17. Why are red tides more frequent in warm coastal waters?
18. Warm conditions favor the rapid growth and reproduction of harmful algae responsible for red tides
19. Differentiate between bioaugmentation and biostimulation

Bioaugmentation : The addition of specific microorganisms to degrade pollutants in contaminated sites. Introduces new microbial strains to break down pollutants that native microbes cannot efficiently degrade.Adding petroleum-degrading bacteria to an oil-contaminated site. Depends on external microbes introduced
Biostimulation : The addition of nutrients or growth stimulants to enhance the activity of native microorganisms. Boosts the existing microbial population to accelerate pollutant degradation. Adding nitrogen or phosphorus to stimulate microbial breakdown of organic waste in water.Depends on native microbes already present in the environment.

20. Explain how heavy metals enter aquatic ecosystem.

Industrial discharge – Effluents from mining, metal processing, battery manufacturing, and electroplating industries release metals like lead, mercury, cadmium, and chromium into water bodies.

Agricultural runoff – Use of pesticides, fertilizers, and sewage sludge containing metals contributes to contamination.

Urban runoff and sewage – Waste from households, plumbing, and urban areas carries metals such as copper, zinc, and nickel into rivers and lakes.

Atmospheric deposition – Metals from air pollution (e.g., from burning coal or smelting) settle on water surfaces or are washed down by rain.

Natural sources – Weathering of rocks and volcanic activity can release small amounts of metals into aquatic environments.

21. Write any two limitations of water quality index

Oversimplification – WQI reduces complex water quality data into a single number, which may hide the details of individual pollutants.
Parameter selection bias – The accuracy of WQI depends on the choice and weighting of parameters, which can vary and affect the interpretation of water quality.

22. Write any two factors which affect the efficiency of bioremediation

Environmental conditions – Factors like temperature, pH, oxygen availability, and moisture influence microbial activity and pollutant degradation.

Nature of the pollutant – The chemical composition, concentration, and bioavailability of the contaminant determine how easily microbes can break it down.

23. How does pesticide accumulation in fish pose risks to human consumers?

Bioaccumulation – Pesticides like DDT and endosulfan accumulate in fish tissues over time, especially in fatty tissues.

Biomagnification – When humans consume these fish, the concentration of pesticides increases in the body, potentially reaching toxic levels.

Health hazards – Long-term exposure can lead to neurological disorders, hormonal disruption, reproductive issues, and even cancer.

Part C
Short Essay Questions
Answer any 4 questions
Each question carries 5 marks (5X4=20)

23. Explain any five disadvantages of phytoremediation
Slow process – Phytoremediation can take months or years to clean up contaminated sites, making it unsuitable for urgent remediation.
Limited to shallow contamination – Plant roots can only reach surface or near-surface pollutants, leaving deeper contaminants untreated.
Species-specific – Only certain plants can tolerate or accumulate specific pollutants, limiting its applicability.
Risk of contaminant transfer – Contaminants can enter the food chain if animals eat the plants used for remediation.
Ineffective for all pollutants – It may not remove highly toxic, volatile, or persistent chemicals efficiently, such as some heavy metals or synthetic organics.
24. Explain how sewage reduces the dissolved oxygen content of water.
Introduction of organic matter – Sewage contains high amounts of organic waste, such as food residues, fecal matter, and detergents.

Microbial decomposition – Aerobic bacteria in water break down this organic matter, consuming large amounts of dissolved oxygen in the process.

Oxygen depletion – As oxygen is used up for decomposition, DO levels drop, creating hypoxic (low-oxygen) conditions.

Impact on aquatic life – Low oxygen can kill fish, invertebrates, and other sensitive organisms, disrupting the ecosystem.

25.Describe oil pollution – sources and its impact on aquatic ecosystems

Oil Pollution refers to the contamination of water bodies with crude oil, petroleum products, or their derivatives, which spread over the water surface and affect aquatic life.

Sources of Oil Pollution:

Oil spills from ships and tankers – Accidental discharge during transportation.

Industrial discharge – Wastewater from petroleum refineries and petrochemical industries.

Urban runoff – Oil from roads, garages, and machinery washed into rivers and seas.

Offshore drilling – Leakage from oil rigs and pipelines.

Impact on Aquatic Ecosystems:

Physical coating – Oil forms a layer on water, blocking sunlight and reducing photosynthesis in aquatic plants.

Toxicity to organisms – Components like PAHs are toxic to fish, invertebrates, and plankton.

Oxygen depletion – Microbial breakdown of oil consumes dissolved oxygen, leading to hypoxia.

Disruption of food chains – Oil bioaccumulates, affecting predators and causing long-term ecological imbalance.

Habitat destruction – Shorelines and wetlands get coated with oil, damaging breeding and feeding grounds.

25. Explain how chemical pollutants affect the quality of water and aquatic life. Alteration of water chemistry – Chemicals like acids, alkalis, and salts change pH, hardness, and conductivity, making water unsuitable for many aquatic organisms.Toxic effects – Heavy metals (lead, mercury, cadmium) and pesticides are toxic, causing disease, reproductive failure, or death in fish and other organisms. Eutrophication – Excess nutrients such as nitrogen and phosphorus from fertilizers promote algal blooms, which block sunlight and reduce oxygen levels when algae decay. Bioaccumulation and biomagnification – Persistent chemicals accumulate in aquatic organisms, leading to higher concentrations in predators, including humans. Disruption of ecosystems – Chemical pollution can eliminate sensitive species, reduce biodiversity, and destabilize food chains.

26.What are the control and preventive measures of oil pollution?
Proper handling and transport of oil – Ensuring safe loading, unloading, and storage to prevent spills from ships, pipelines, and tankers.
Oil spill containment – Using booms, barriers, and skimmers to contain and remove oil from water surfaces immediately after a spill.
Use of dispersants and absorbents – Chemicals or materials that break down or absorb oil, reducing its spread and impact.
Regular maintenance of equipment – Preventing leaks from refineries, offshore rigs, and machinery through proper inspection and repair.
Wastewater treatment – Treating industrial effluents containing oil before discharging into water bodies.
Legislation and monitoring – Enforcing strict environmental regulations and continuous monitoring to minimize accidental discharges.
28.Discuss the biological productivity differences between eutrophic and
oligotrophic lakes.

	Eutrophic	**Oligotrophic**
Nutrient levels	High in nitrogen and phosphorus	Low in nutrients
Primary productivity	Very high – abundant algae and aquatic plants	Low – limited growth of algae and plants
Dissolved oxygen	Often low in deeper layers due to decomposition of organic matter	High throughout the water column
Water clarity	Turbid or murky due to algal blooms	Clear, with high transparency
Biodiversity	Supports large populations of fish and invertebrates adapted to high nutrient conditions, but may have oxygen stress	Supports fewer organisms, but water is oxygen-rich and stable
Organic matter accumulation	High – leads to faster sedimentation and possible eutrophication	Low – sediments accumulate slowly

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